Course: CSTE 5131 (Machine Learning)
MS: 2022 - 23

Instructor/Teacher:

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Textbook & Lecture Notes:

There is no required text for the course. Course notes will be posted on google classroom. But I recommend the following additional reading:

- **Textbook:** "*Data Mining: Concepts and Techniques*", J. Han, M. Kamber, and J. Pei, Morgan Kaufmann (3rd Edition), "*Machine Learning*" by Tom Mitchell, McGraw-Hill
- **Lecture notes by Andrew Ng:** http://cs229.stanford.edu/lectures-spring2022/main_notes.pdf
- **Strongly Recommended:** Class lectures (available after each class)

Class (≈23)	Topics
1, 2	Introduction to Machine Learning (ML), overview of DM and Big Data,
3, 4	Data Preprocessing/ Evaluation Metric
5, 6, 7, 8, 9, 10	Supervised Learning: Classification/Prediction/Regression
11, 12	Neural Networks
13, 14	Ensemble Learning
15, 16	Learning Theory
17, 18, 19	Unsupervised Learning/Cluster Analysis
20, 21	Introduction to Deep Learning
22, 23	Visualization

Evaluation:

- Exam - 70%
- Assignment/Class Test/Presentation: 25%
- Attendance: 5%

Web Resource Examples:

- **General Data Mining (DM)/Machine Learning (ML) Site:** http://www.kdnuggets.com
    - A collection of DM/ML publications, software/tools, data repositories, companies, etc.
- **Data Mining and Machine Learning Software/Tools/Programming Language – Weka**: http://www.cs.waikato.ac.nz/ml/weka/
    - A collection of machine learning algorithms for solving real-world data mining problems. It is written in Java and runs on almost any platform.

  R: https://cloud.r-project.org/
  Python: https://www.python.org/
- **Database Repository:**
  http://www.ics.uci.edu/~mlearn/MLRepository.html
  https://www.kaggle.com/
  https://datasetsearch.research.google.com/
  https://visualdata.io/discovery